

Regulation of Plasma Glucose

Charles P. Scott, Ph.D.

October 6th, 2026

Learning Objectives

1. Summarize the mechanism of glucose sensing by pancreatic beta cells.
2. Compare the insulin sensitivities of lipolysis, glucose production, and glucose disposal.
3. Outline maturation of peptide hormones using insulin as an example. Explain the source of C-peptide and the clinical utility of C-peptide assays.
4. Describe the normal physiological counter-regulation to insulin-induced hypoglycemia.
5. Relate the structures, functions and regulation of glucagon, GIP and GLPs and describe how GIP and GLPs enhance insulin secretion.

Fed vs. fasting state

Alternating use of glucose and fatty acids

Fed state

Too much glucose

Use a lot, store some

(Store fatty acids)



Fasting state

Too little glucose

Make glucose from stores and amino acids

Decrease glucose use

Increase use of fatty acids

Learning Objectives

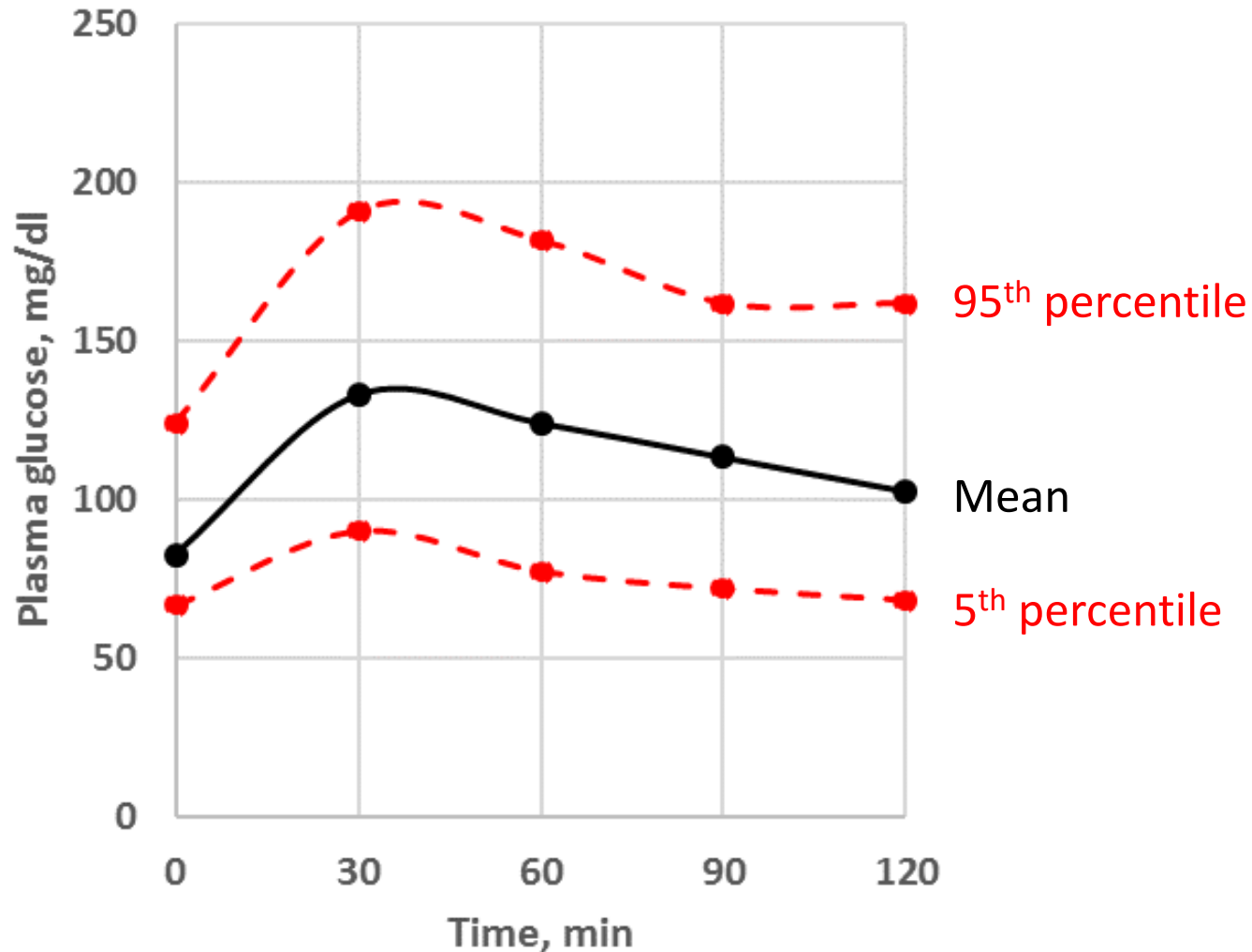
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ScholarRX 39394.5

A 28-year-old man is found nauseated, delirious, and weak by a rescue team after becoming lost during a mountain hike. The patient had access to water but had no food for 4 days. He is tremulous and oriented to person only. His serum glucose concentration is 60 mg/dL. Which of the following is most likely in this patient's islet β cells?

- A. Decreased activation of voltage-gated calcium channels
- B. Downregulation of GLUT-2 transporters
- C. Increased protein synthesis
- D. Inhibited oxidative metabolism
- E. Voltage-gated calcium channel activation with intracellular calcium accumulation

Oral Glucose Tolerance Test (75g)

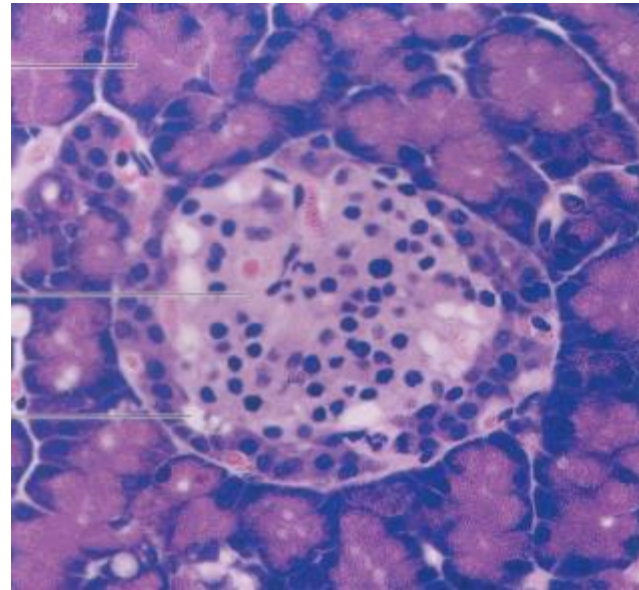
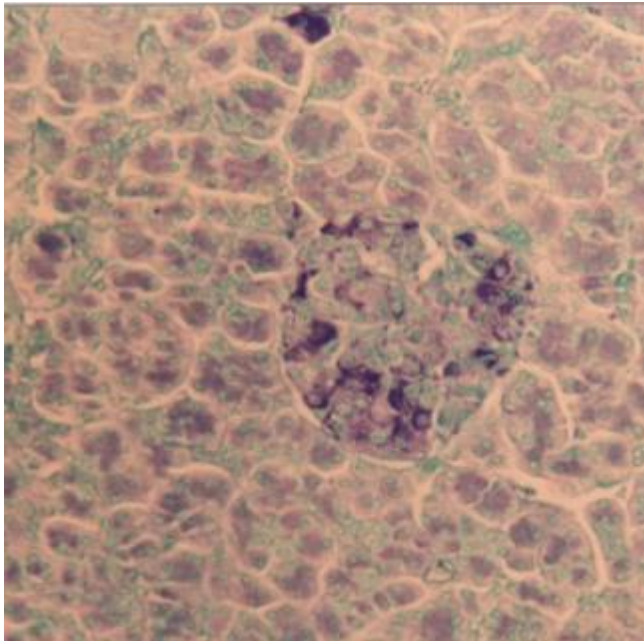


Pancreatic islets

= Islets/Islands of Langerhans

~1% of weight of pancreas

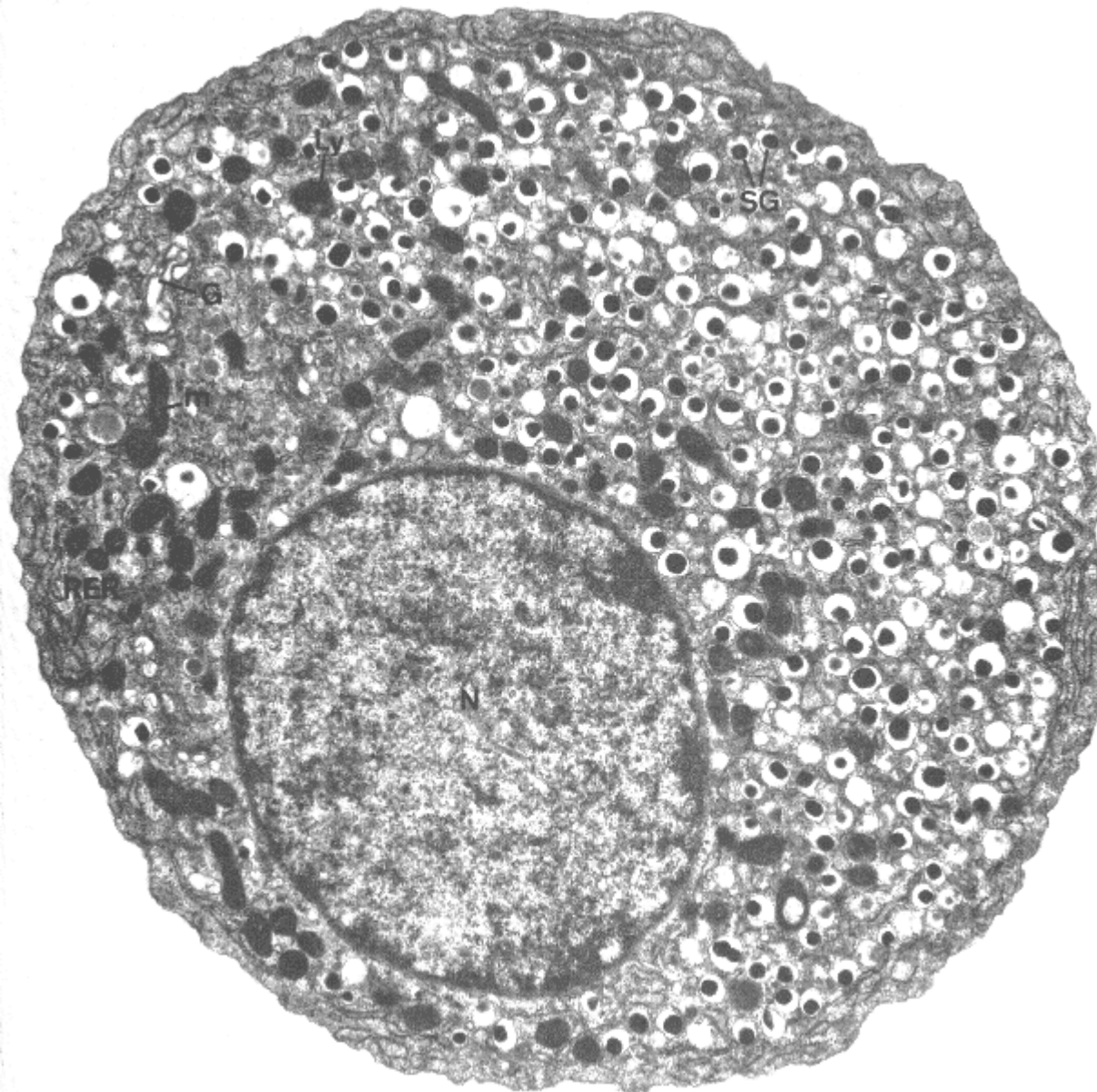
Secrete hormones that regulate fuel metabolism



The islets of Langerhans

<u>Cell type</u>	<u>Hormone synthesized</u>
α -cells	Glucagon
β -cells	Insulin
δ -cells	Somatostatin
F-cells	Pancreatic polypeptide

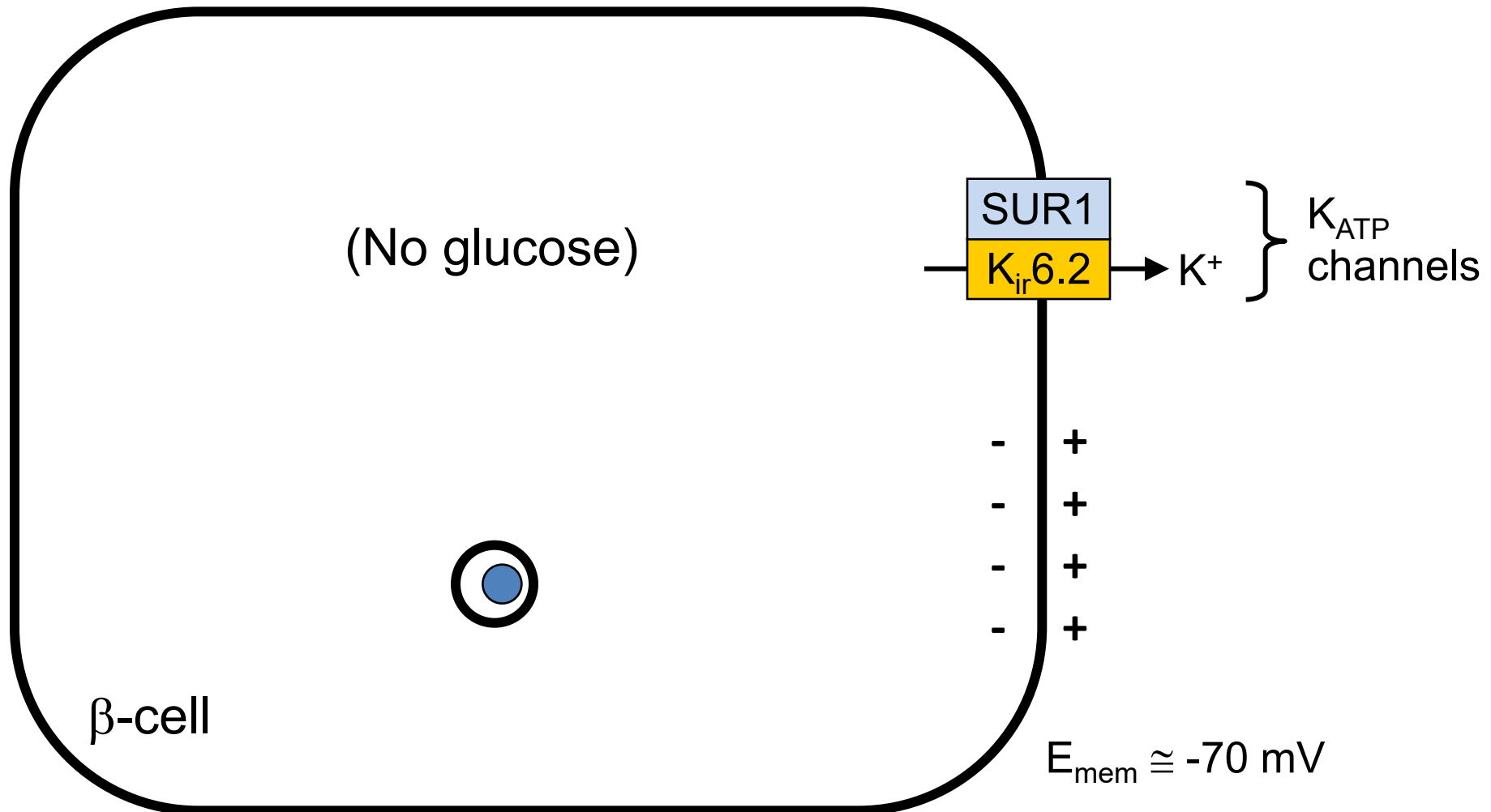
Electron micrograph of a pancreatic β -cell



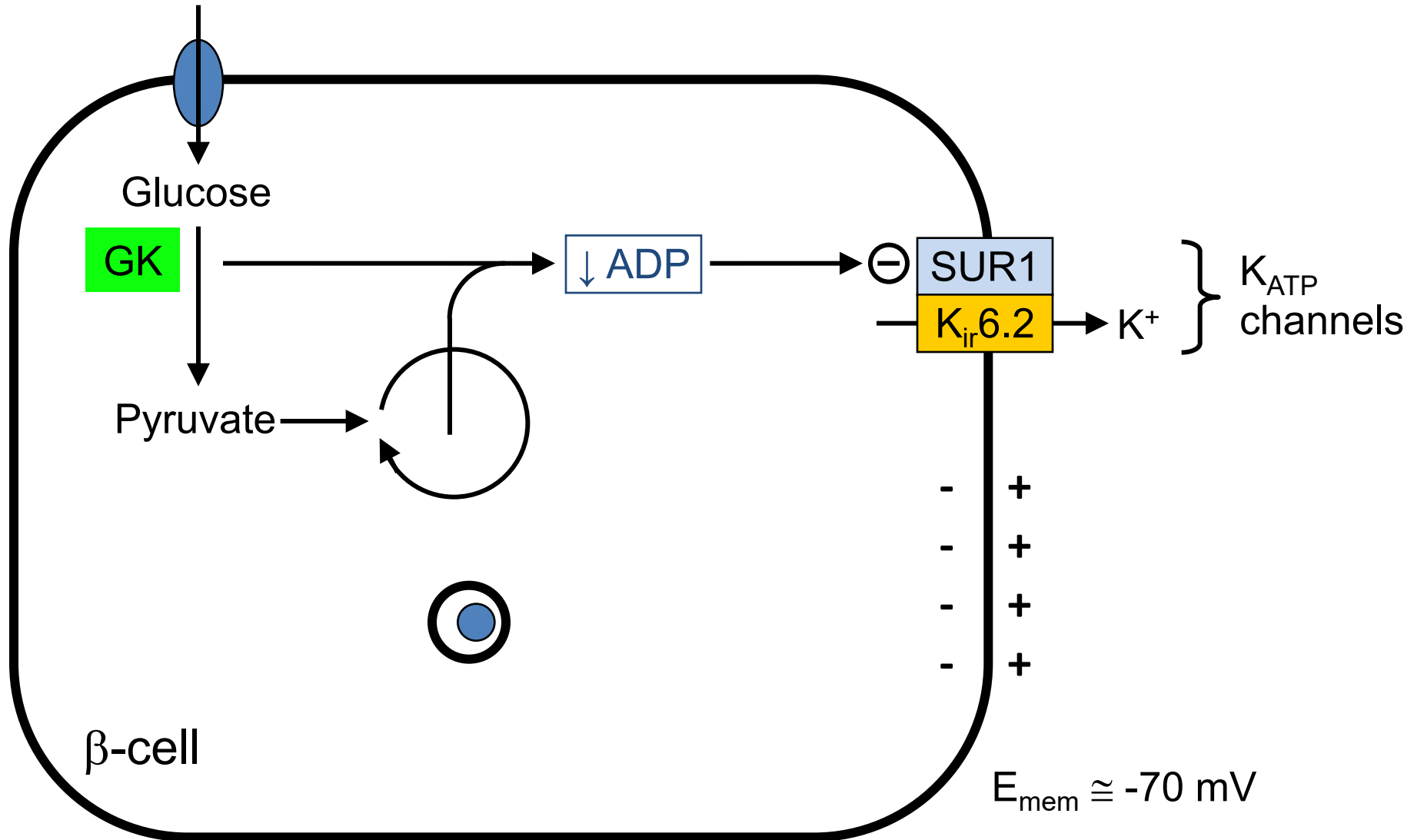
β -cells store
about a
week's
worth of
insulin.

From: L. Orci, Diabetes 31: 538-565, 1982

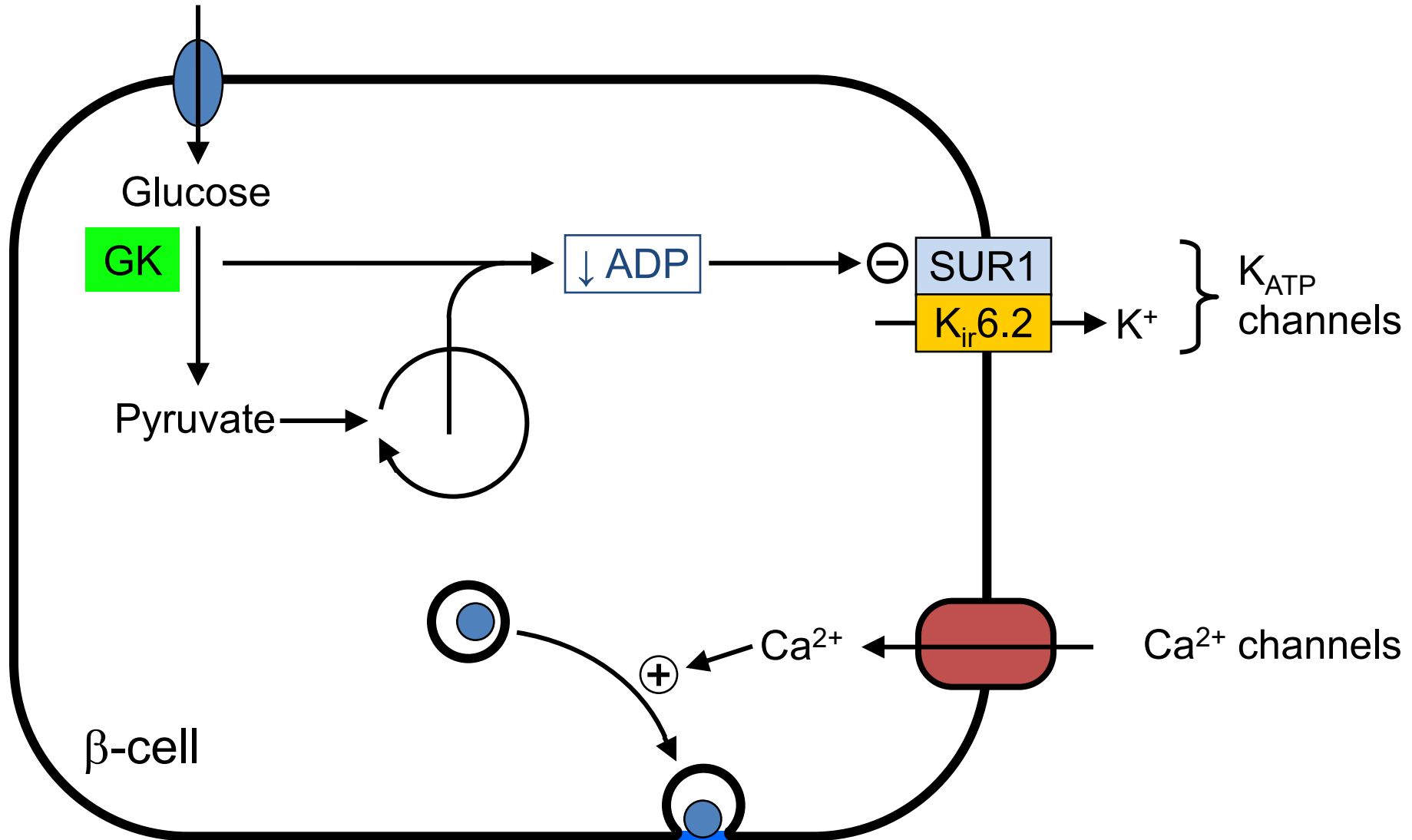
How does glucose elicit insulin secretion?



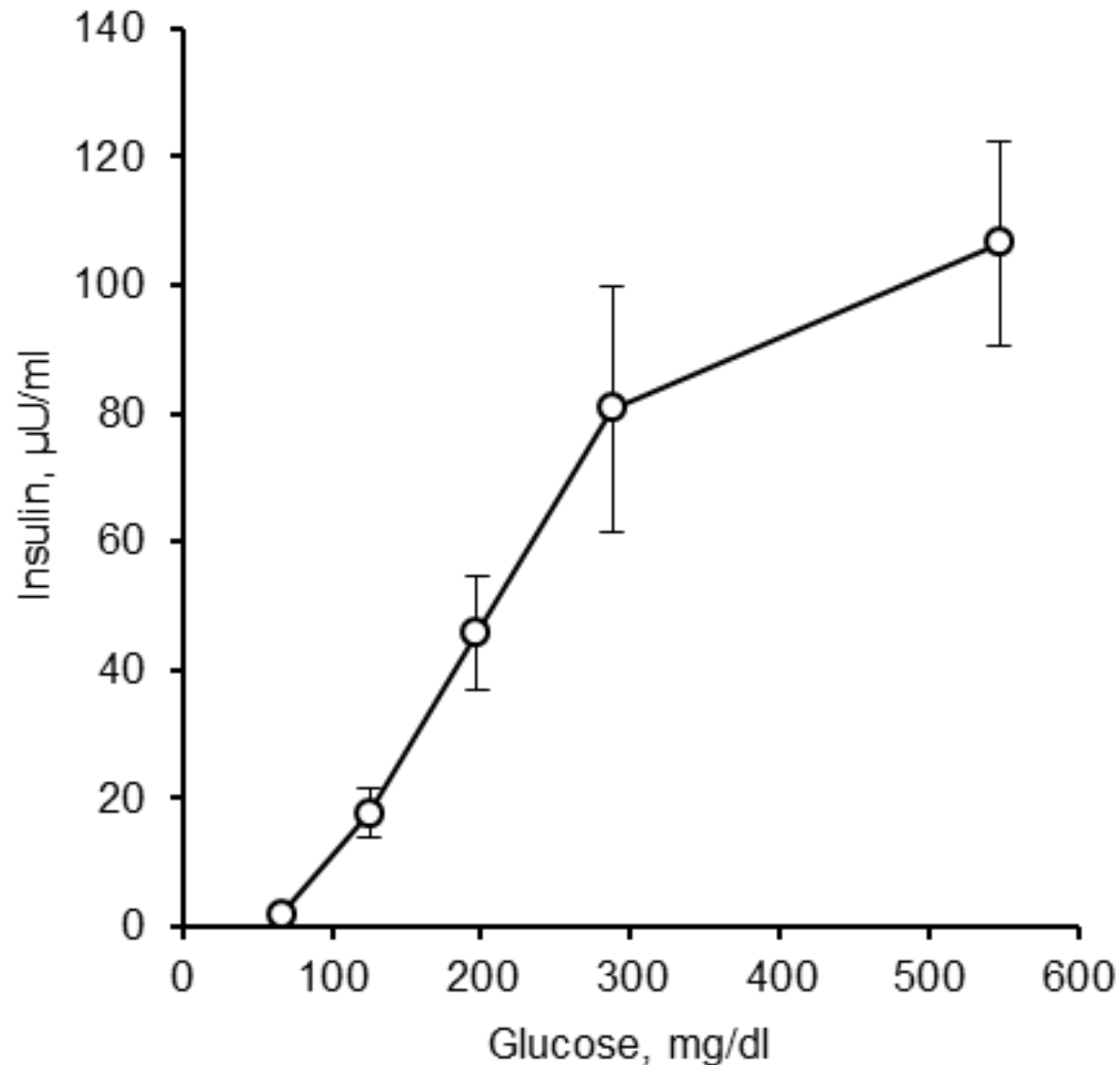
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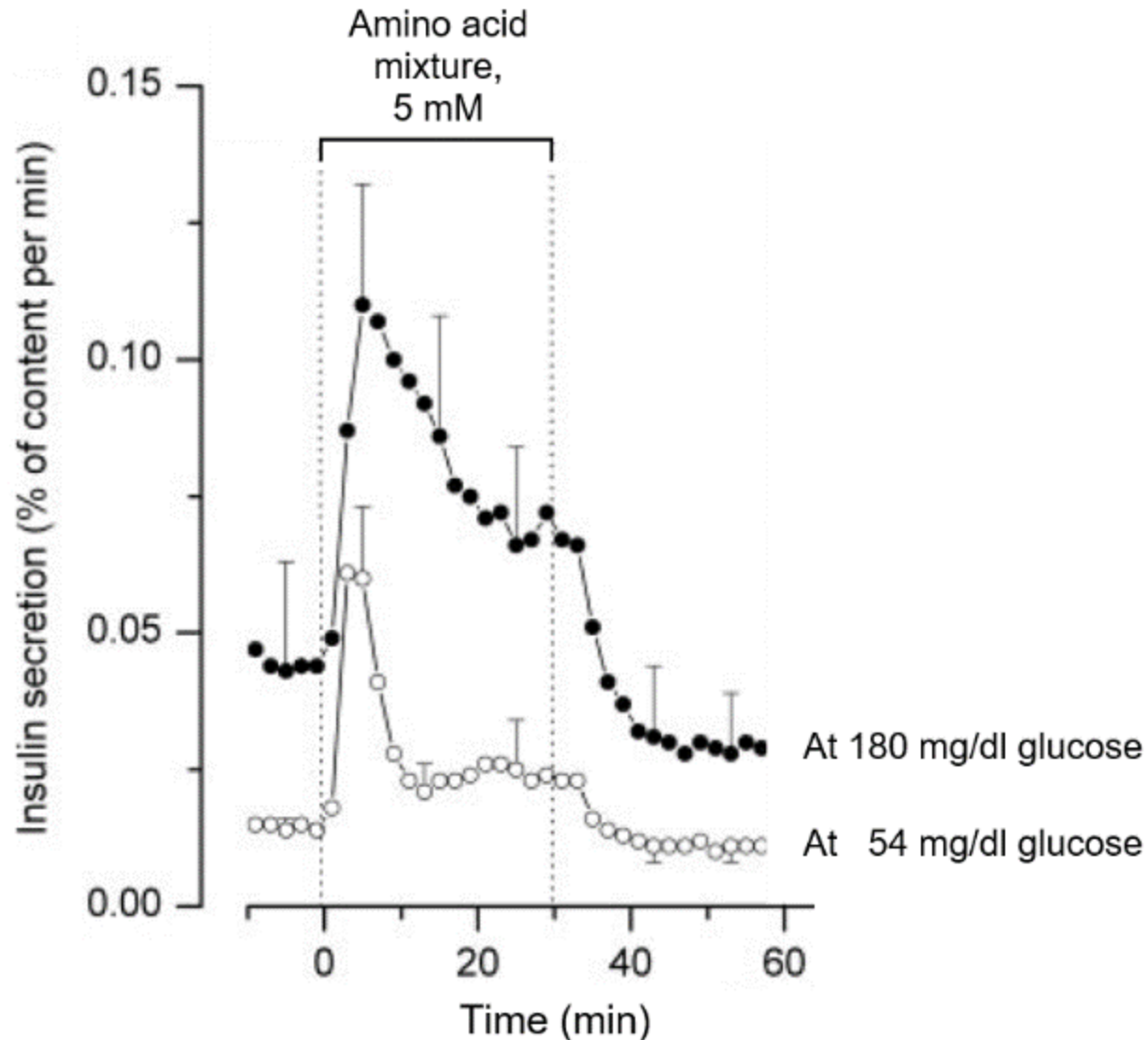
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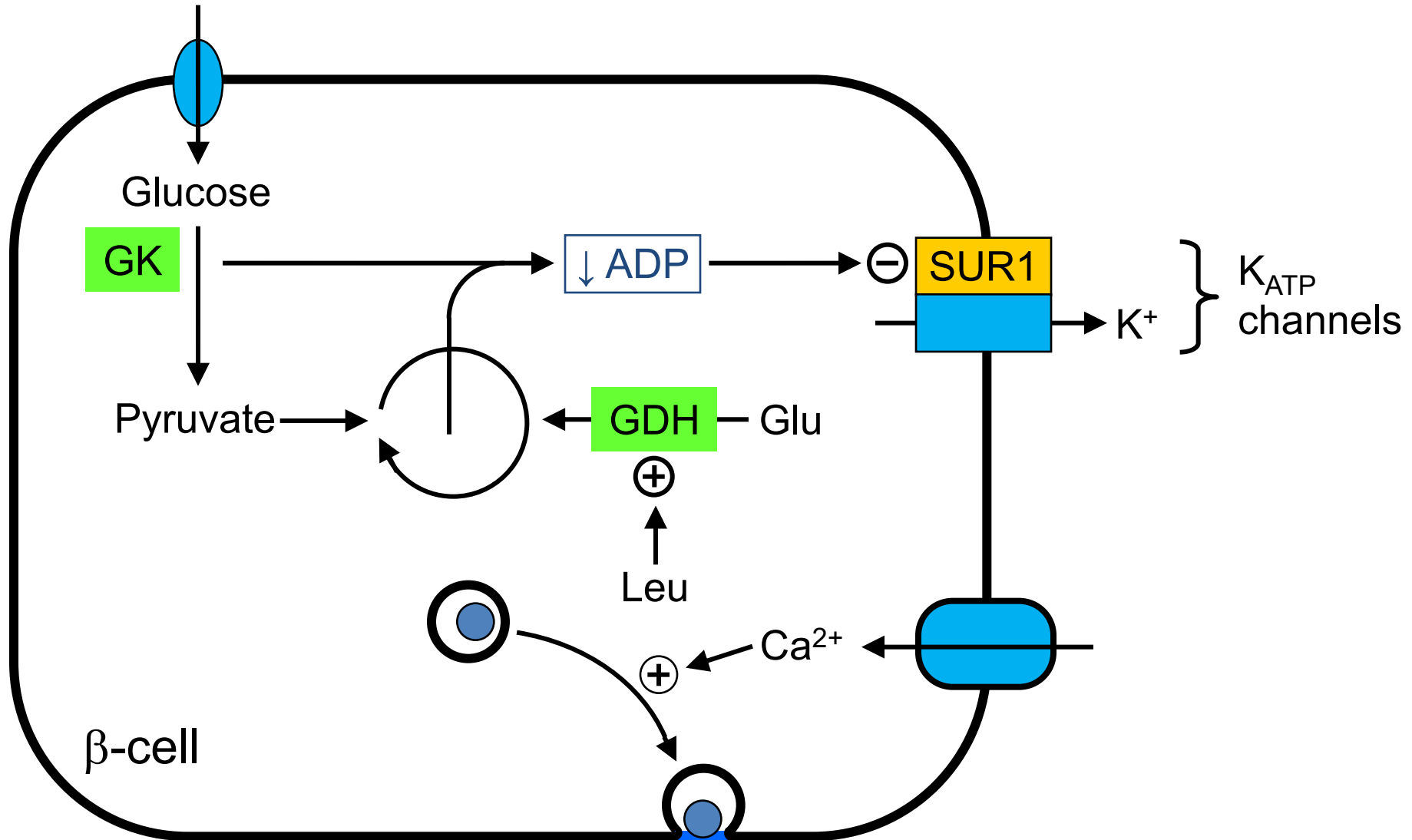
Glucose stimulates insulin secretion



Amino acids enhance glucose-induced insulin secretion



How do amino acids elicit insulin secretion?



ScholarRX 39394.5

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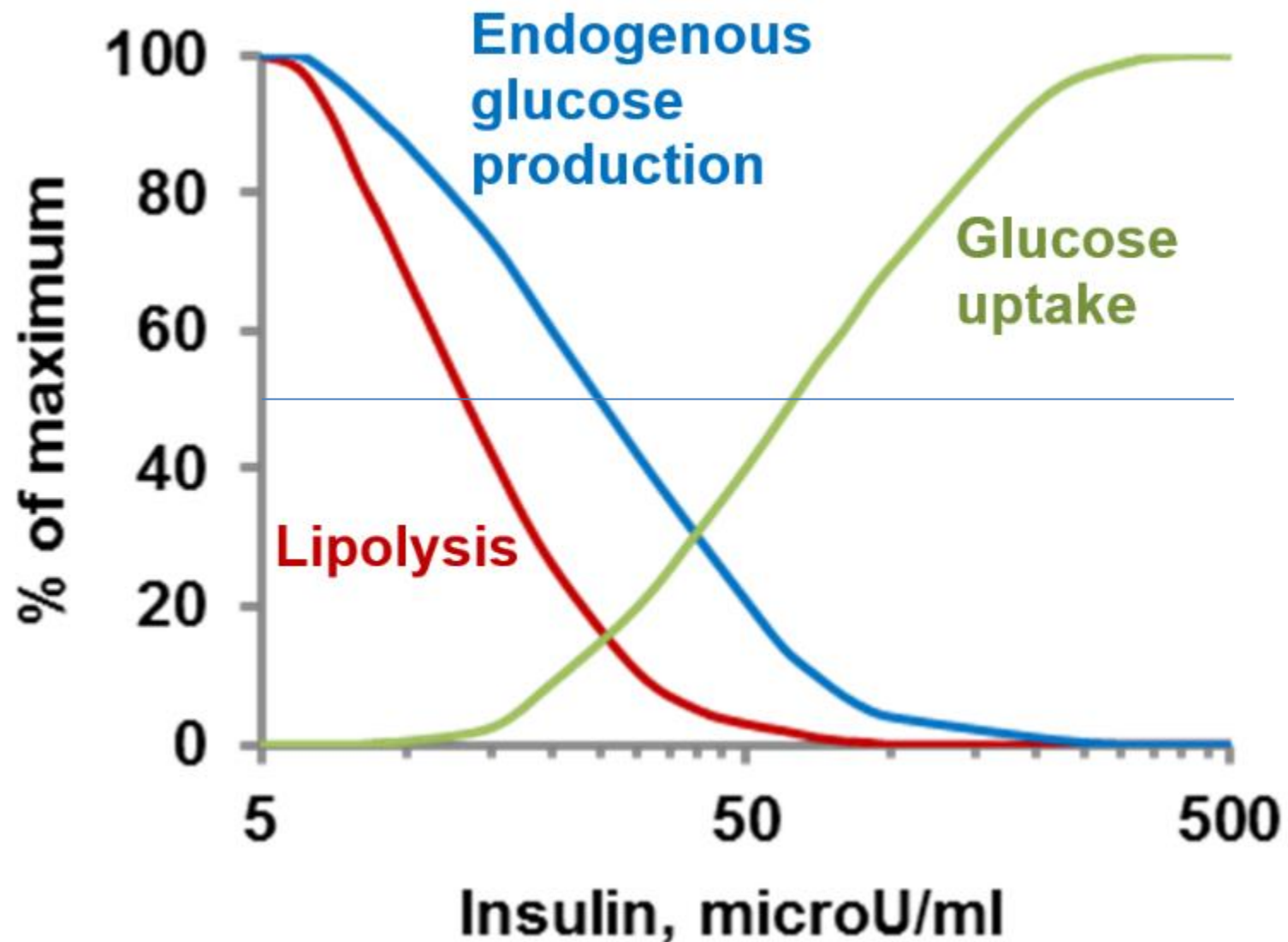
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ScholarRx 1405.18

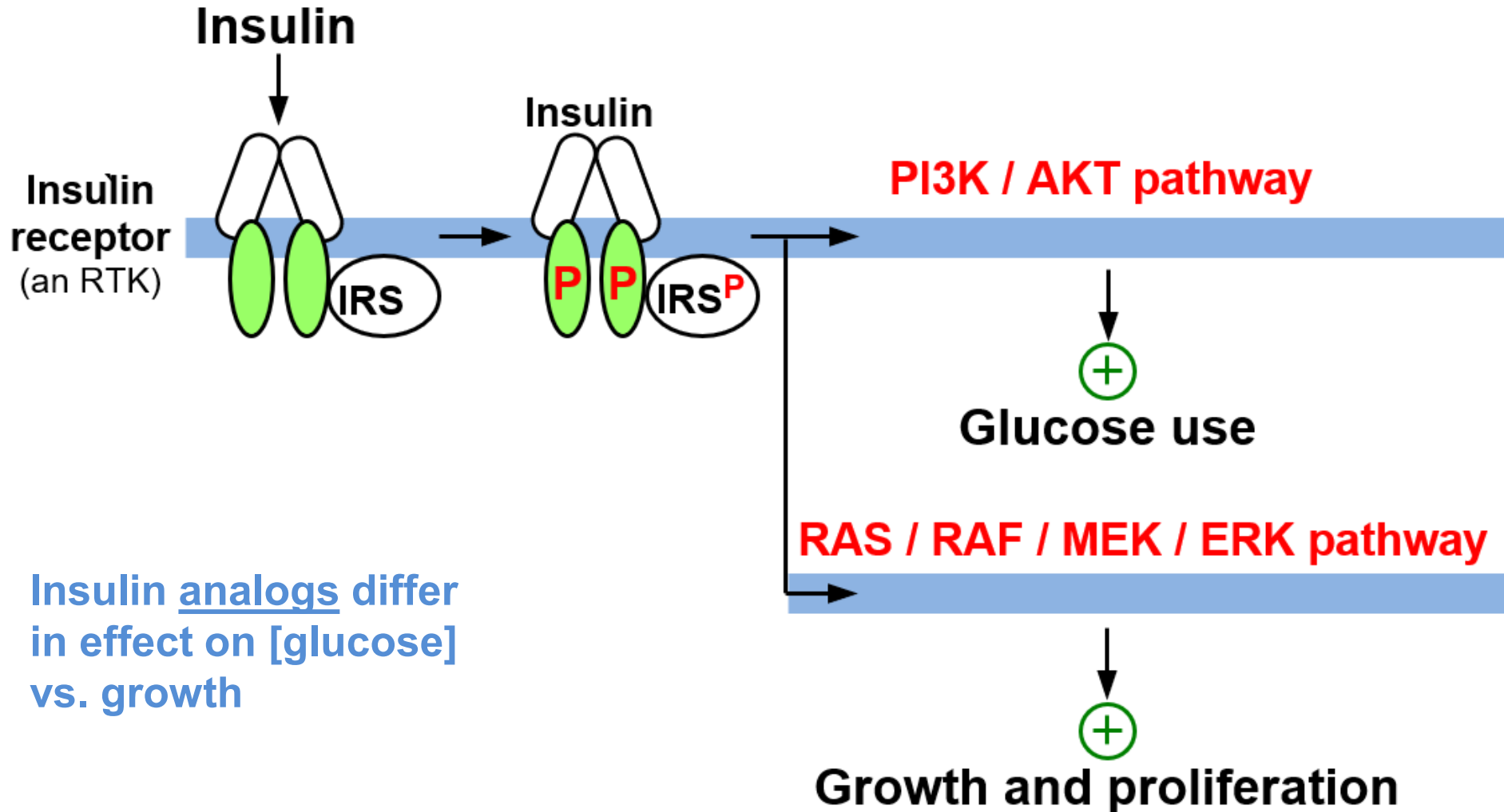
A 74-year-old man is brought to the emergency department because of lightheadedness, weakness, increased thirst, and increased urination. Symptoms began 3 days ago when he started taking prednisone to treat arthritis. Laboratory studies show a glucose level of 294 mg/dL. The rate of which of the following enzymatic conversions is most likely increased in this patient?

- A. Glucose to glucose-6-phosphate
- B. Malate to oxalate
- C. Oxaloacetate to phosphoenolpyruvate
- D. Phosphoenolpyruvate to pyruvate
- E. Pyruvate to acetyl-CoA

Insulin sensitivity of pathways of metabolism



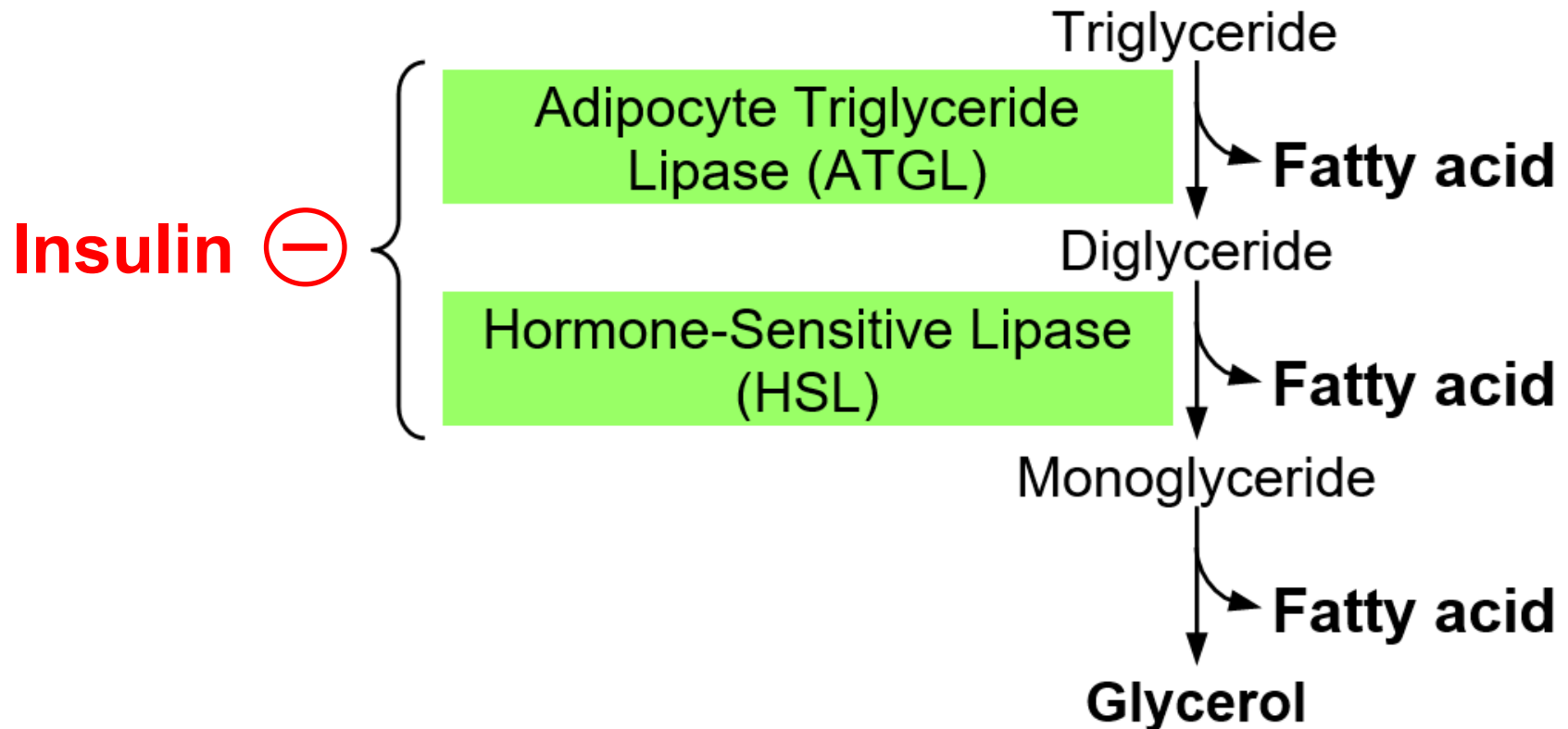
Two insulin receptor signaling pathways



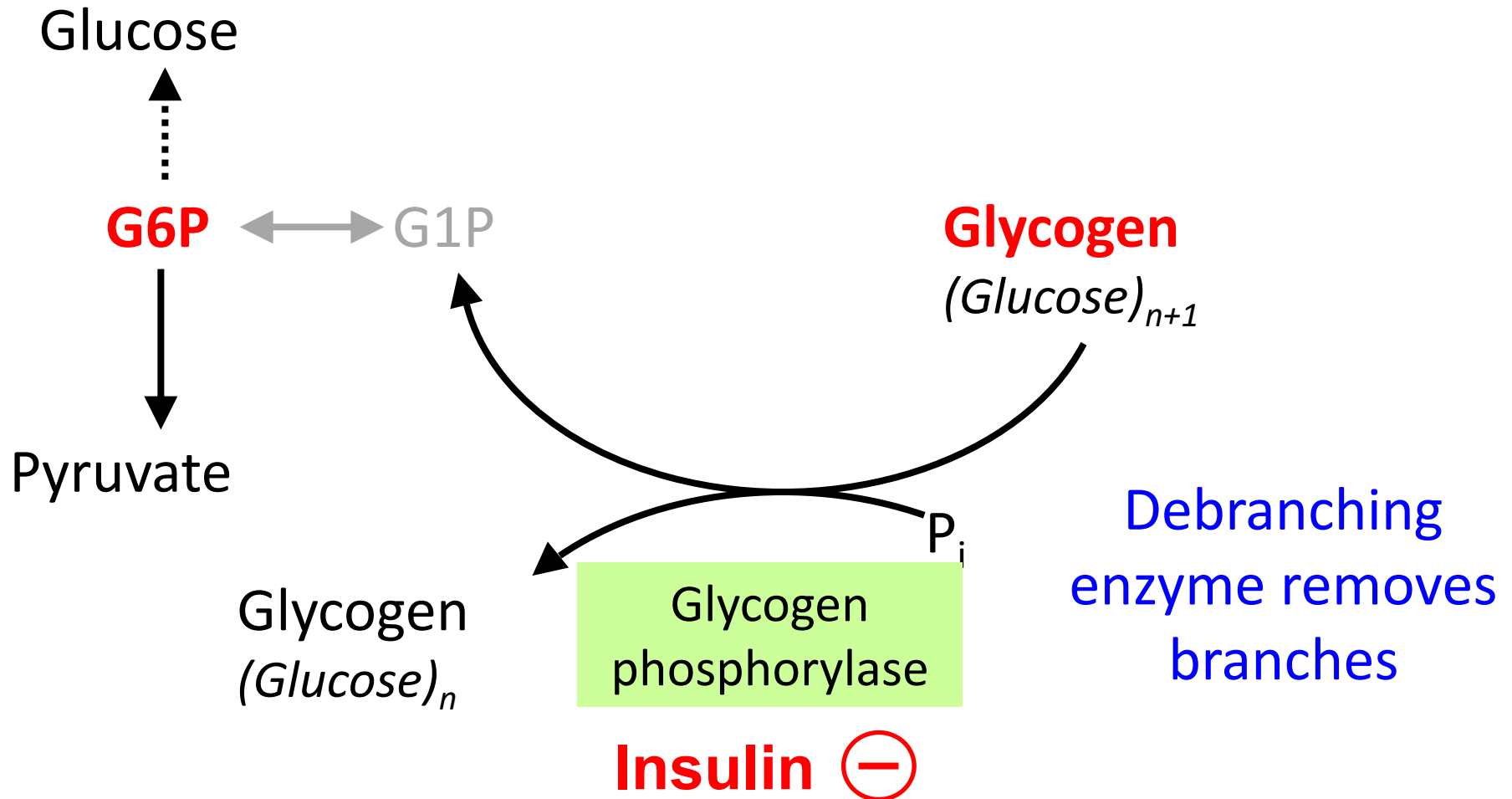
Lipolysis

Definition of term:

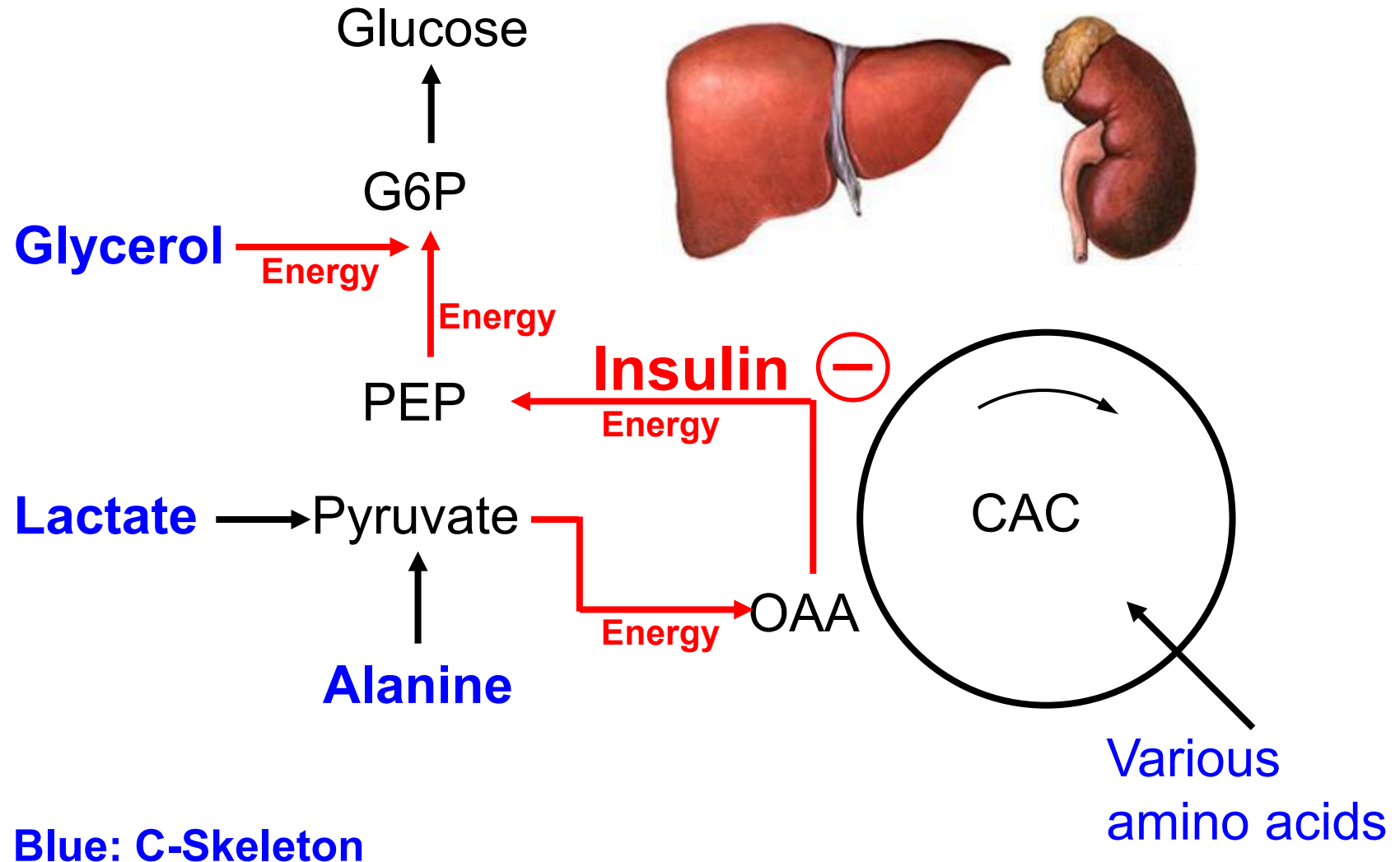
Hydrolysis of triglycerides stored inside **adipocytes**



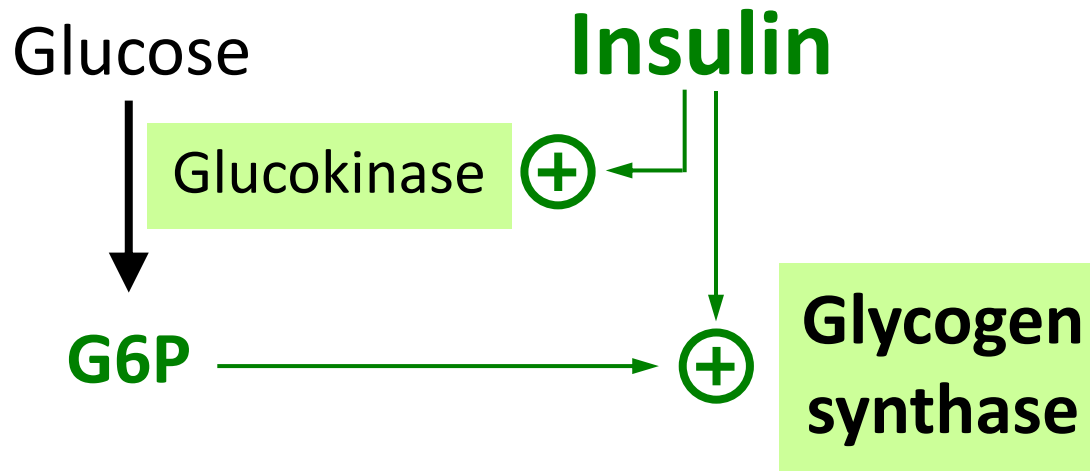
Glycogen degradation



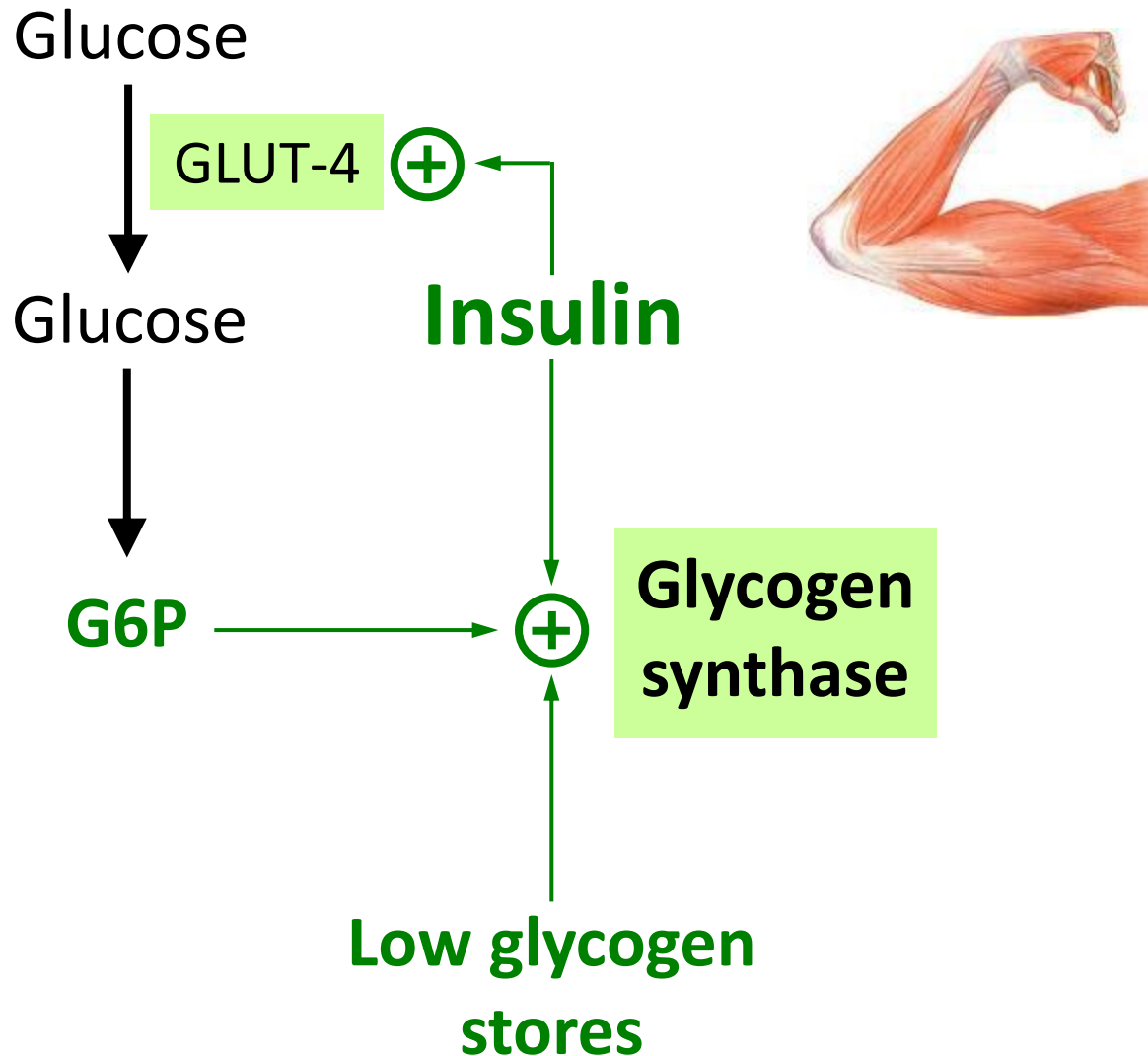
Gluconeogenesis



Control of glycogen synthesis in liver



Control of glycogen synthesis in muscle



ScholarRx 1405.18

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Summary so far...

Insulin lowers plasma glucose by:

- Blocking endogenous energy production (via inhibition of lipolysis, glycogenolysis and gluconeogenesis), thereby forcing utilization of serum glucose
- Stimulating glycolysis (via activation of glucokinase)
- Stimulating glucose sequestration (via membrane insertion of GLUT4 in skeletal and cardiac muscle and adipose tissue)
- Stimulating glycogen synthesis in liver and muscle

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ScholarRx 2682.22

A 30-year-old woman comes to the clinic because of diaphoresis, lethargy, and episodes of confusion for 3 months. She reports these symptoms often occur in the morning before breakfast and sometimes after prolonged periods without eating. She takes no medications. Laboratory studies show a low serum glucose level and elevated C-peptide level. This patient's symptoms are most likely due to activation of which of the following receptors?

- A. α 1-adrenergic receptor
- B. Hormonal G protein-coupled receptor
- C. Intranuclear receptor
- D. M_3 -muscarinic receptor
- E. Tyrosine kinase receptor

Typical synthesis of peptide hormones

Hormone mRNA



Preprohormone



Prohormone



Hormone

Pre-sequence (signal sequence)
delivers peptide to E.R. translocon

Enzyme in E.R. removes pre-
sequence

Prohormone travels through Golgi

Enzymes in secretory vesicles
convert prohormone to hormone

Synthesis of insulin

Insulin mRNA



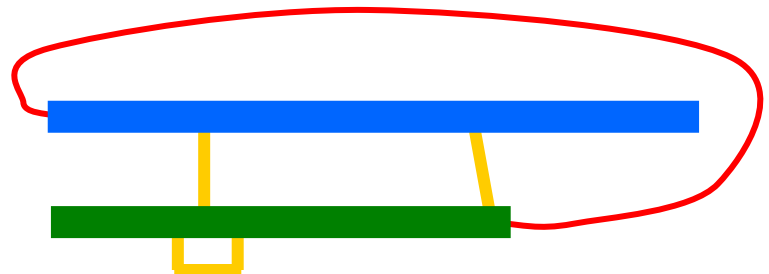
Preproinsulin



Proinsulin



Insulin



(Insulin) C-peptide

Is a measure of endogenous insulin secretion

Commercial recombinant insulin does not contain C-peptide

Examples of questions to answer:

Does this patient have an insulinoma, or are they injecting insulin?

How much insulin does this patient with diabetes still secrete?

ScholarRx 2682.22

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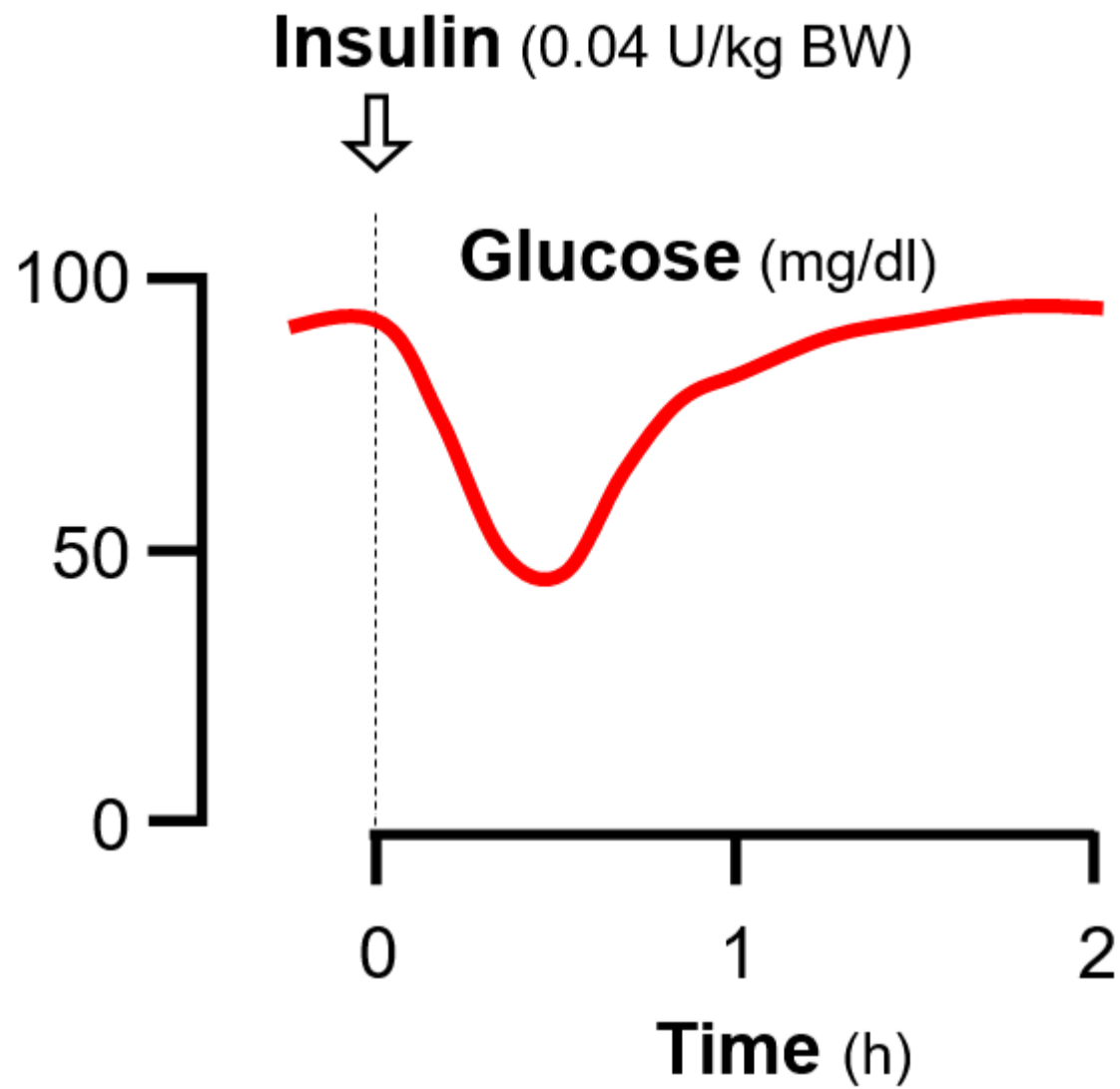
Learning Objectives

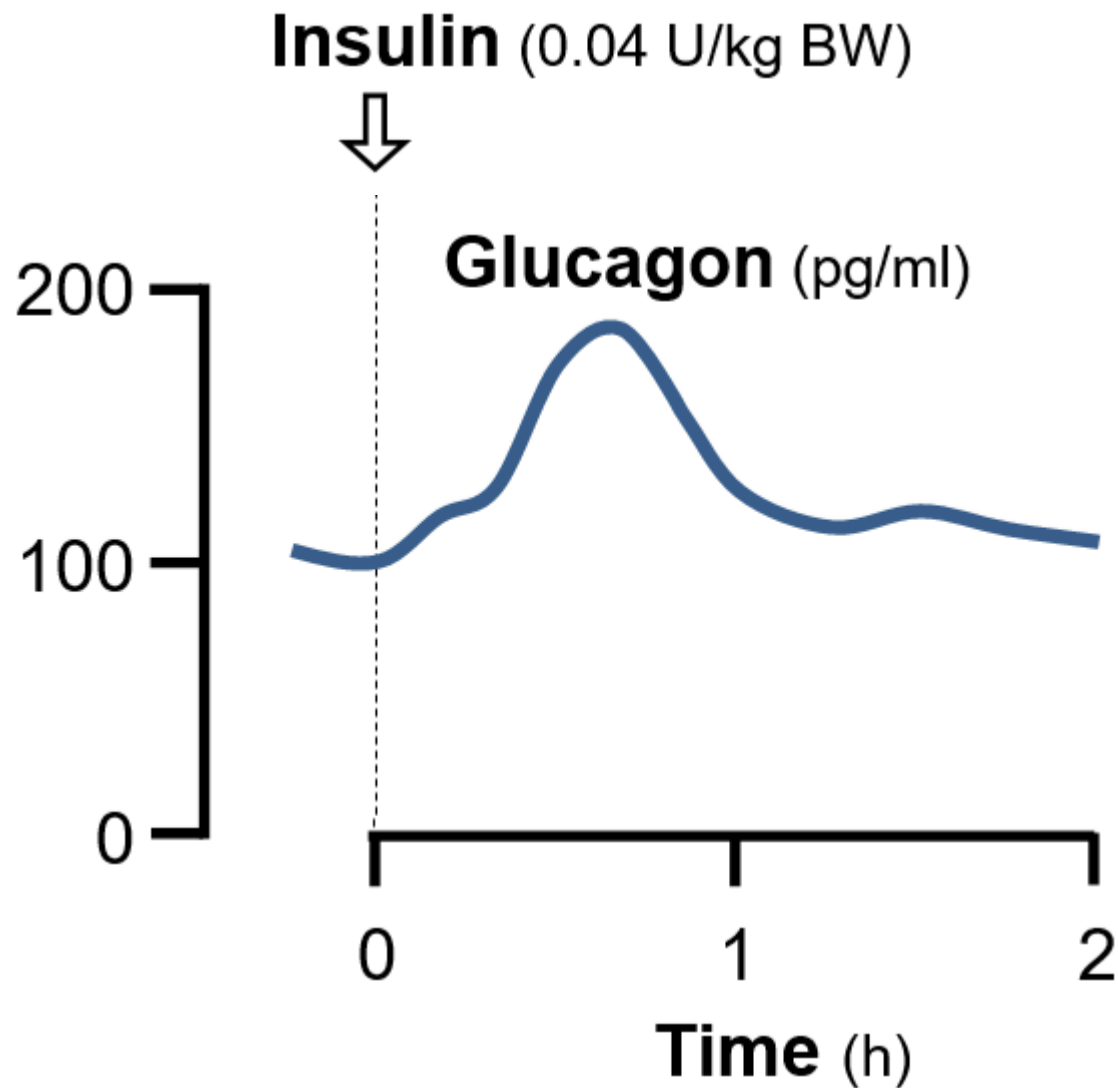
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ScholarRx 3850.11

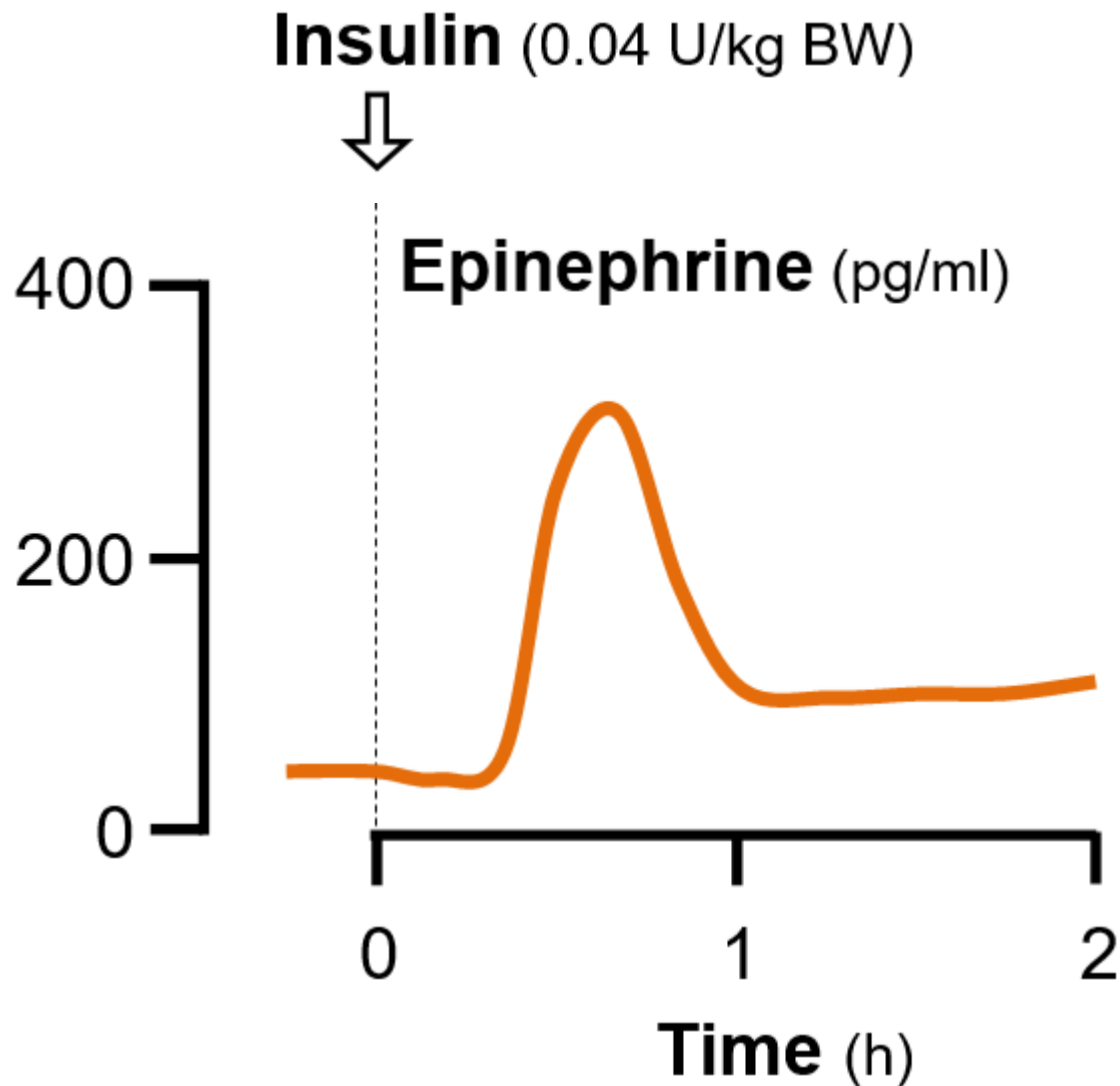
A researcher creates a knockout mouse with an inactivated receptor. The receptor normally binds a hormone whose secretion from the pancreatic islets increases during times of starvation. Which type of receptor is most likely functioning abnormally in this knockout mouse?

- A. Cytosolic receptor
- B. G protein-coupled receptor
- C. Ion channel-linked receptor
- D. Receptor serine/threonine kinase
- E. Receptor tyrosine kinase

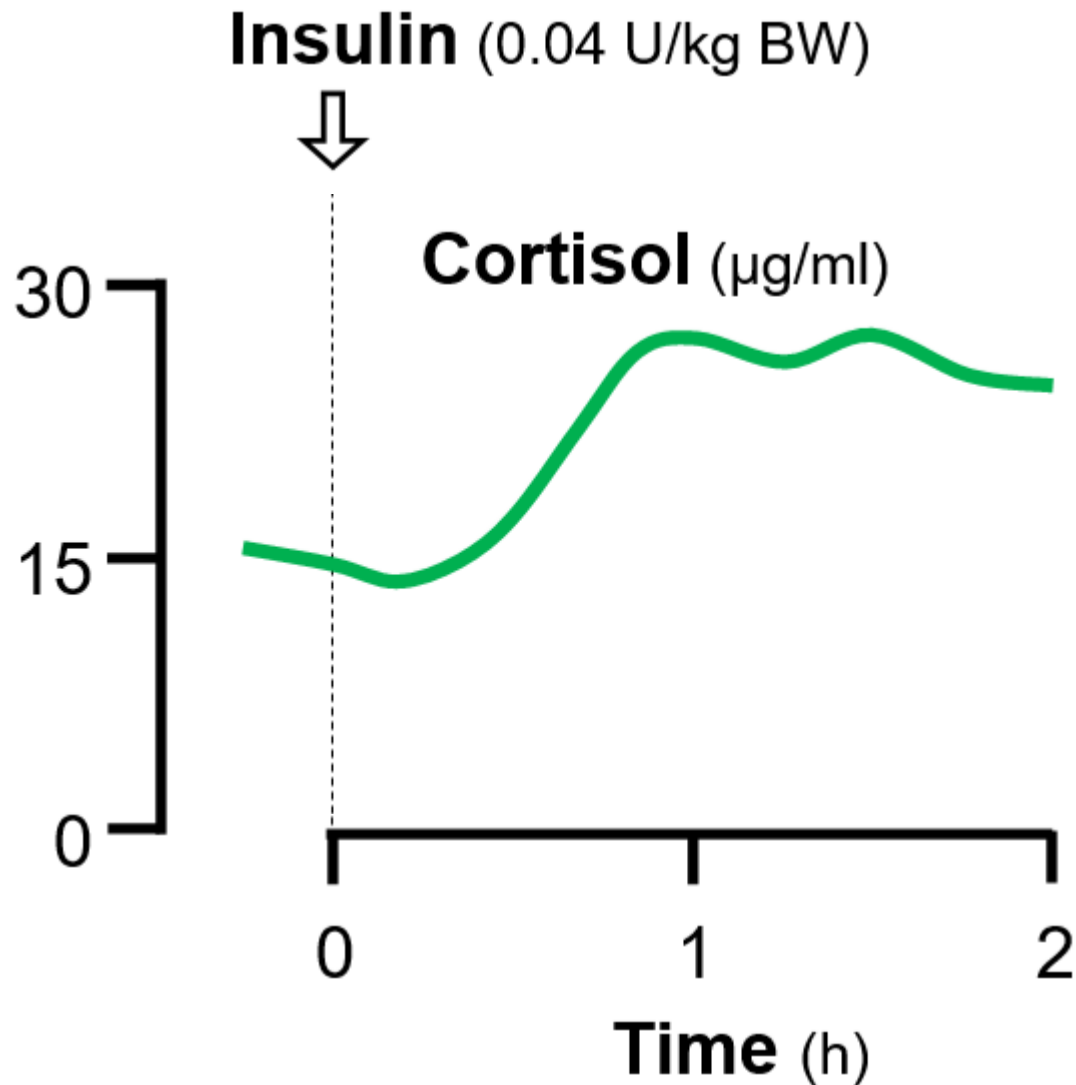




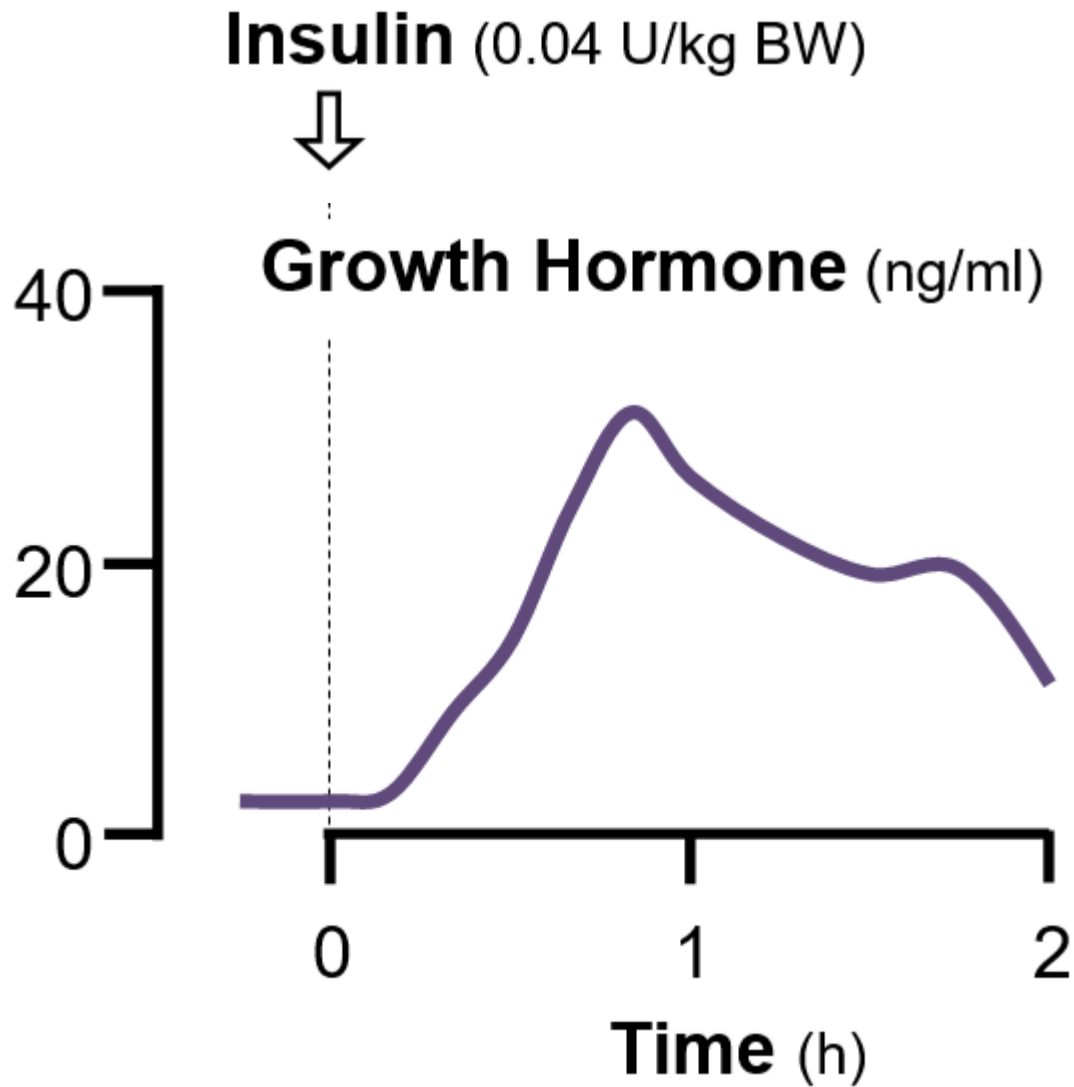
Glucagon stimulates glycogenolysis and gluconeogenesis



Epinephrine stimulates glycogenolysis, gluconeogenesis, and lipolysis



Cortisol stimulates proteolysis and expression of aminotransferases, thus favoring gluconeogenesis



Growth hormone stimulates glycogenolysis, and gluconeogenesis

Control of glycogenolysis in liver



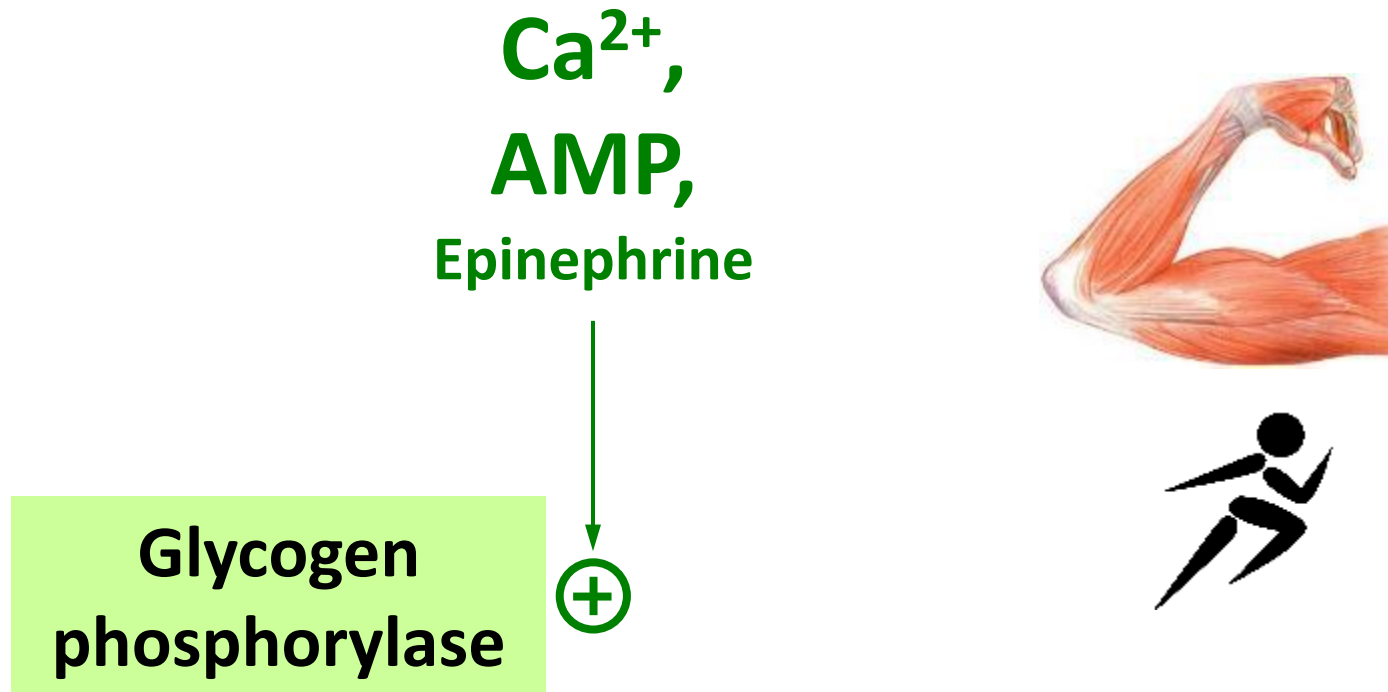
Glucagon,
Epinephrine



Glycogen
phosphorylase



Control of glycogenolysis in muscle



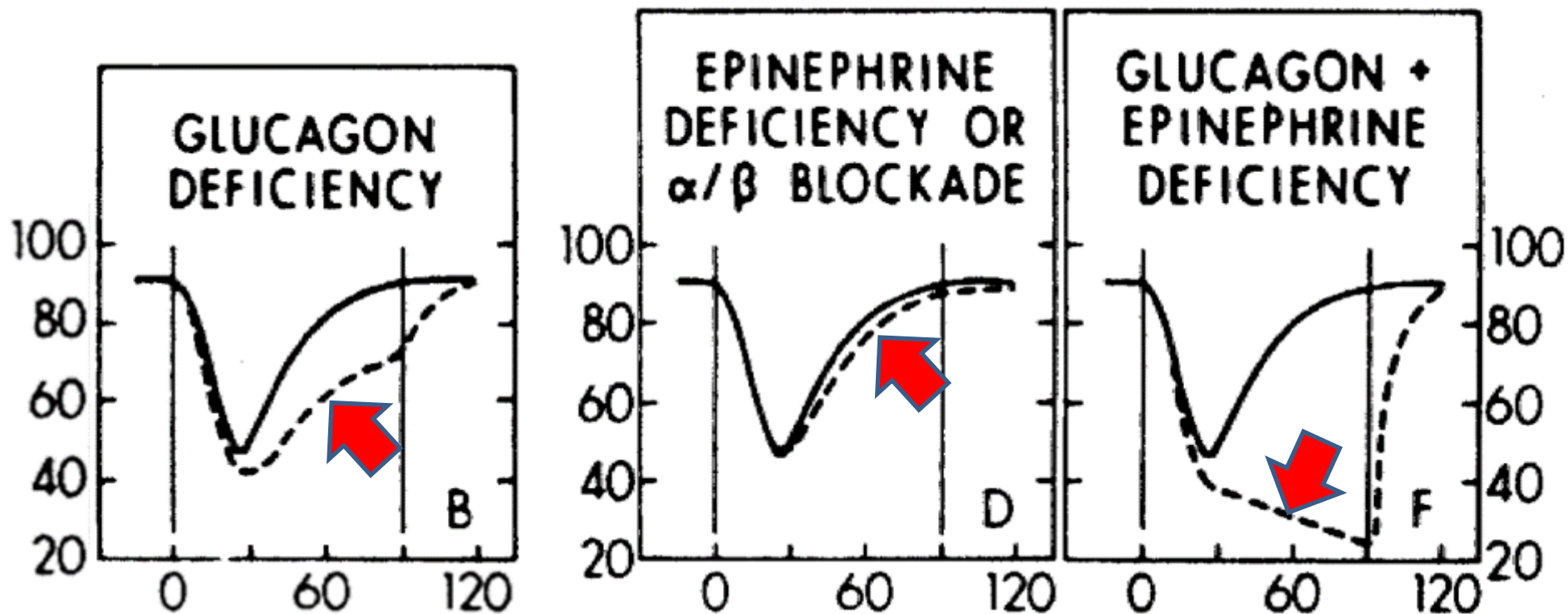
How is GNG regulated?

Long-term at 1st enzyme that catalyzes an
“irreversible” reaction: PEPCK (OAA → PEP)

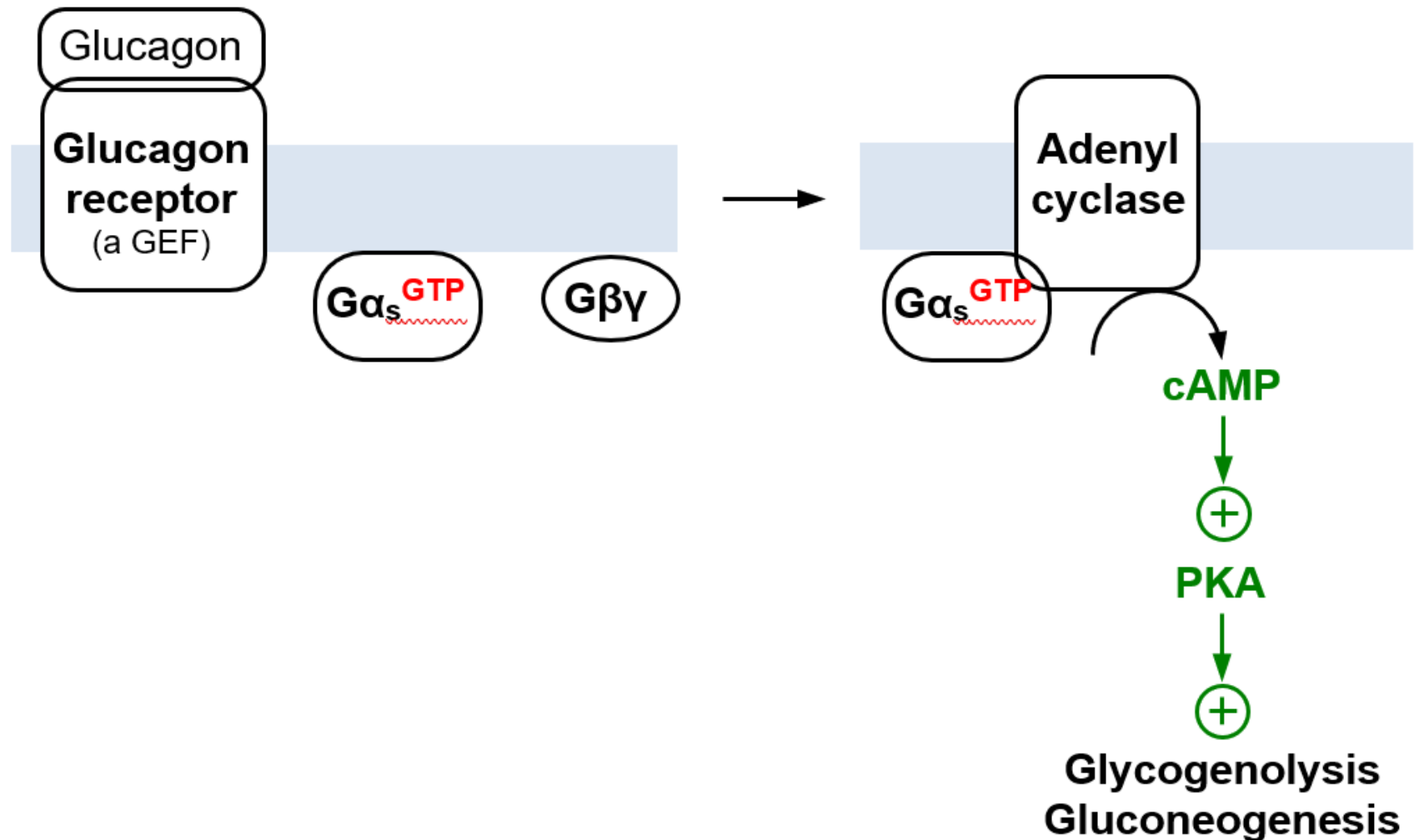
Stimulated by **epinephrine, glucagon**

Inhibited by **insulin, AMP, AMPK**

Which counterregulatory hormones are essential?



Glucagon receptor signaling



ScholarRx 3850.11

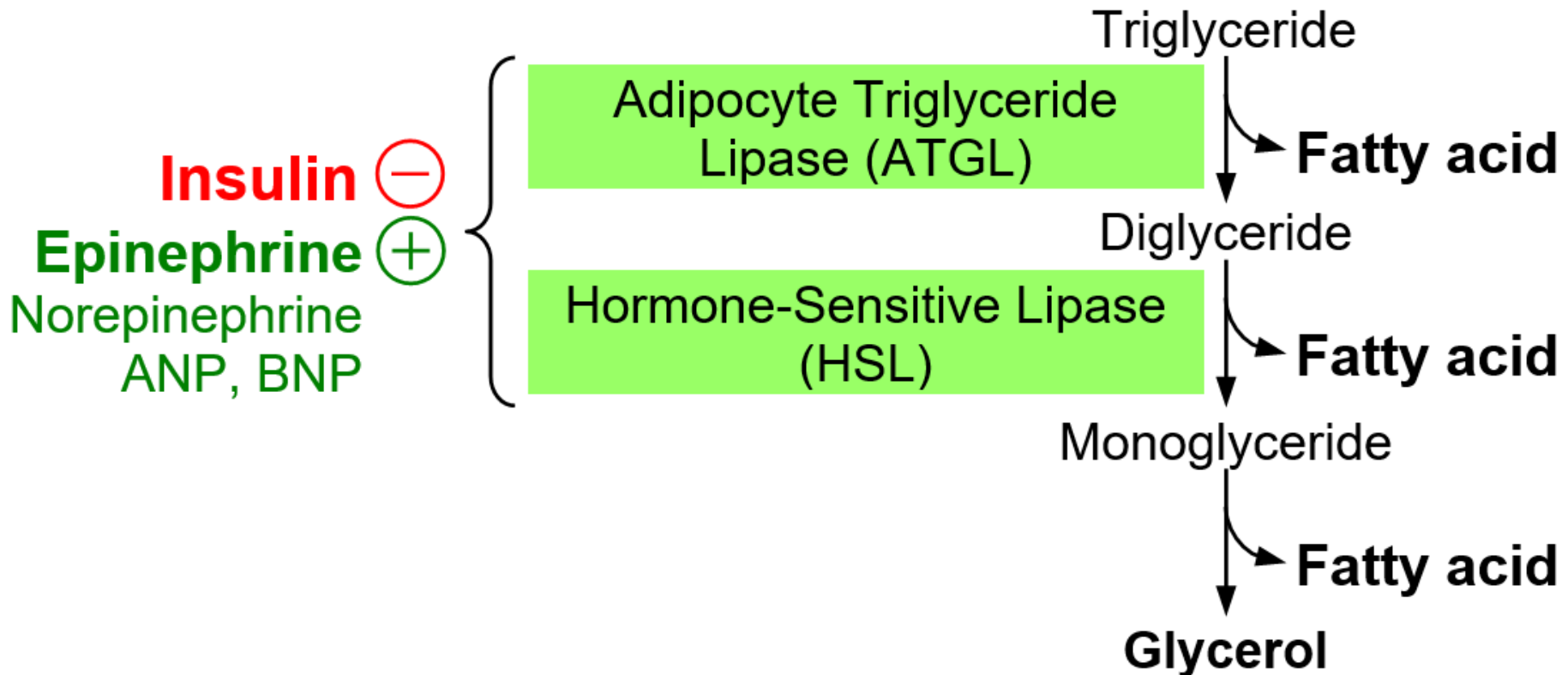
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Lipolysis

Definition of term:

Hydrolysis of triglycerides stored inside **adipocytes**



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ScholarRx 2800.18

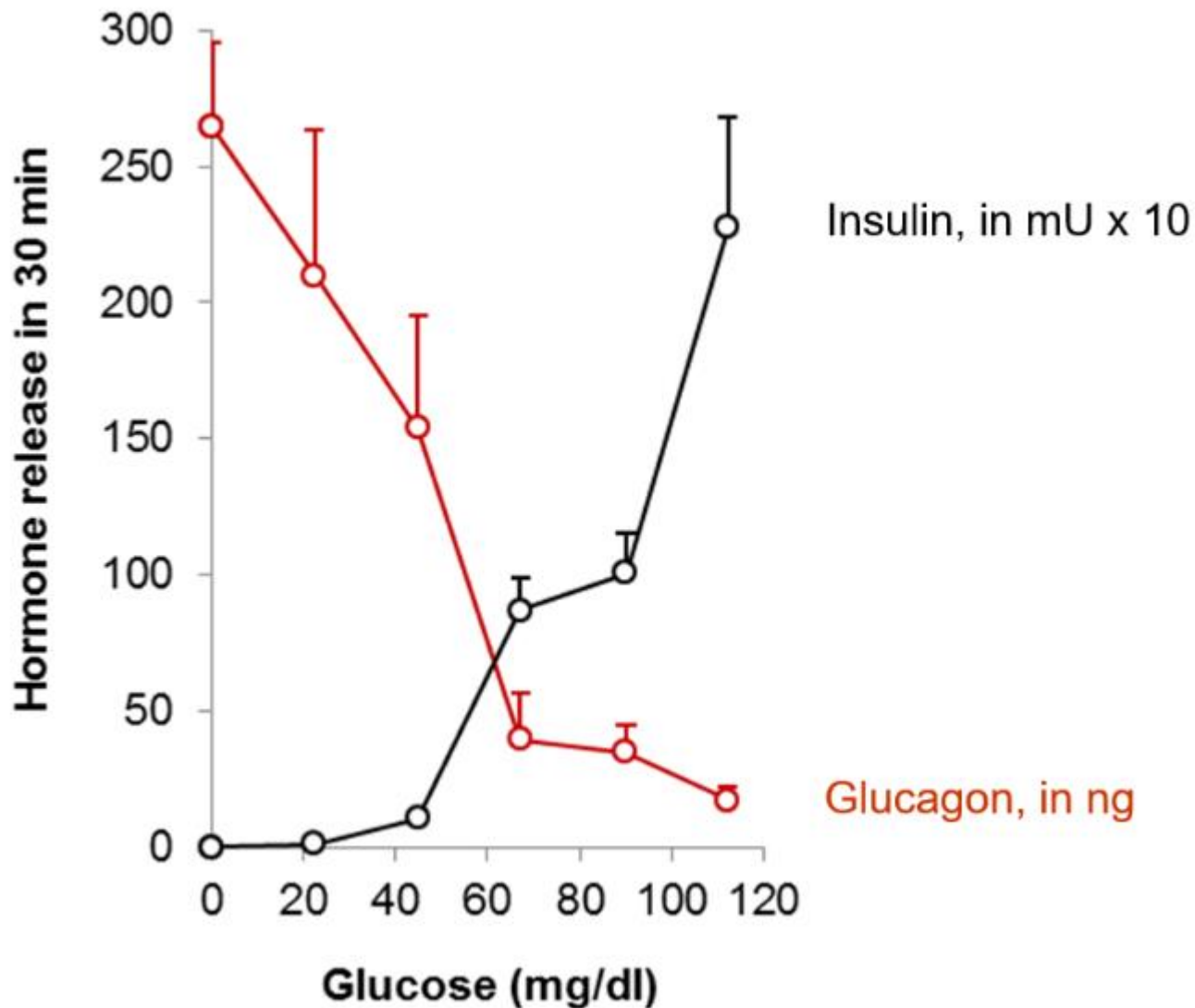
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- A. Epinephrine
- B. Growth hormone
- C. Incretins
- D. Insulin-like growth factor 1
- E. Ketone bodies
- F. Norepinephrine

Glucagon secretion

Hypoglycemia stimulates glucagon secretion

Insulin and Glucagon are secreted concurrently



Synthesis of glucagon in α -cells

Glucagon mRNA



Preproglucagon



Proglucagon



Glucagon

Glucagon



GLP-1



GLP-2



K-cells in intestine produce GIP

GIP = gastric inhibitory peptide

= glucose-dependent
insulinotropic polypeptide

GIP and GLPs are **incretins**



L-cells in intestine produce GLPs

GLPs = glucagon-like peptides (1 & 2)

Synthesis of GLP-1 and GLP-2 in L-cells

Glucagon mRNA



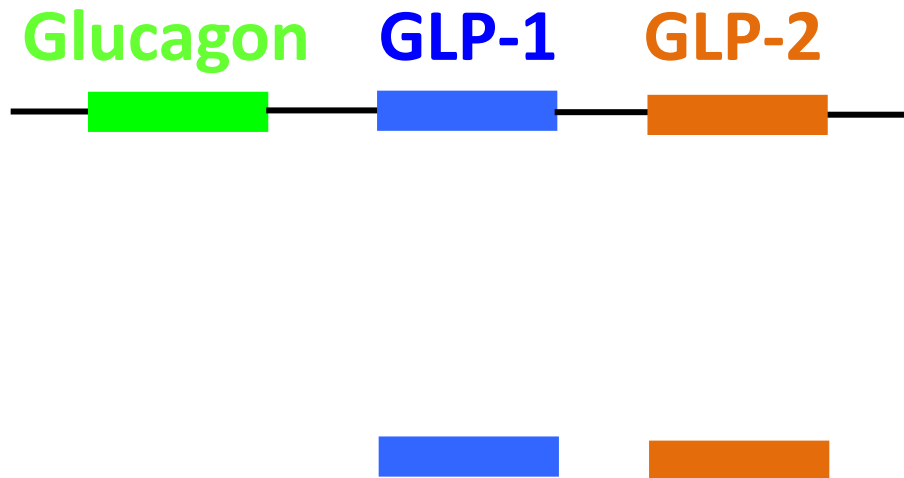
Preproglucagon



Proglucagon



GLP-1 and GLP-2



GIP, GLP and glucagon receptor signaling

GIP, GLP and glucagon receptors are **GPCRs**

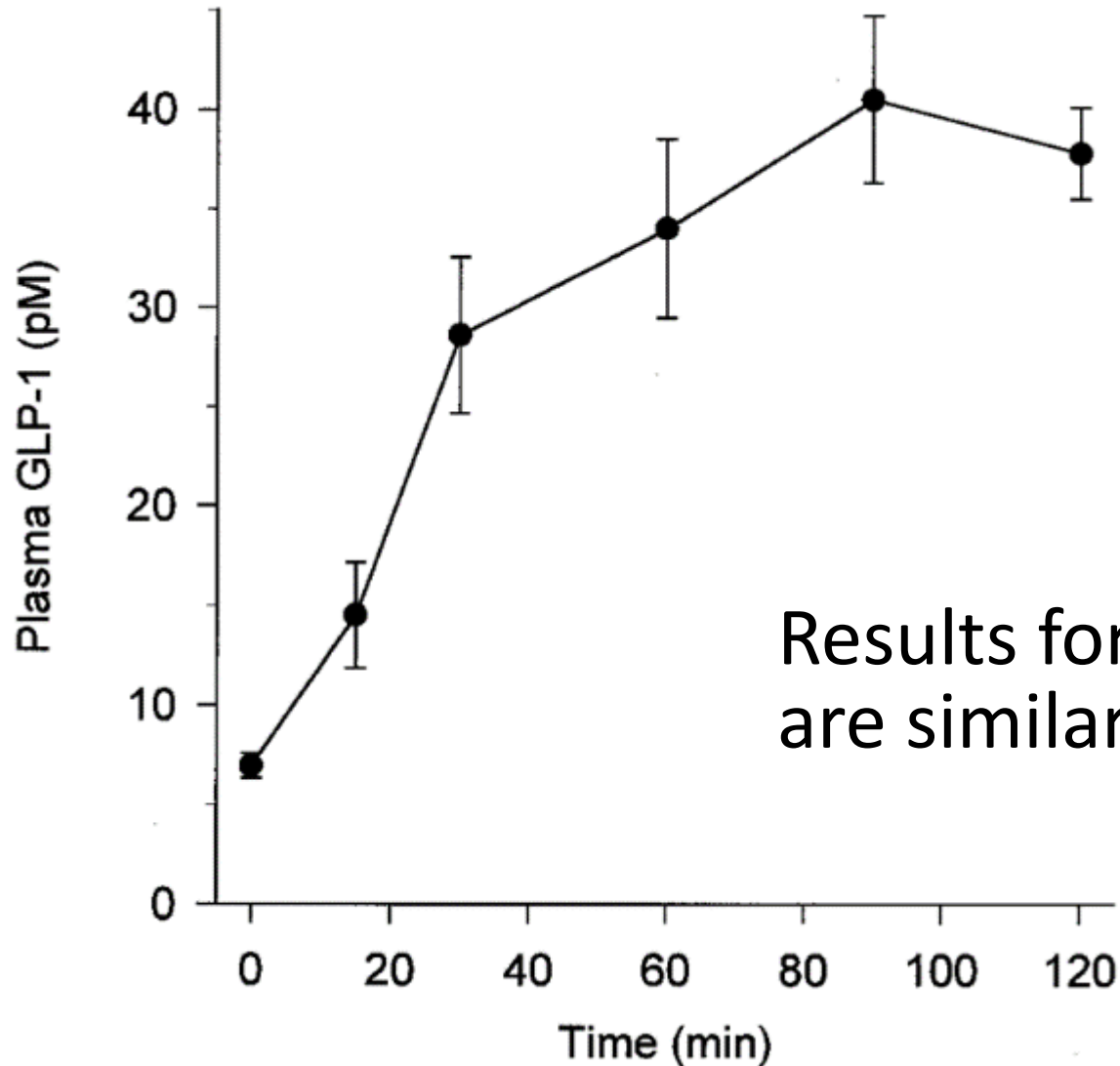
Each couples to **G_{as}**, generating cAMP as a second messenger to activate PKA

Glucagon receptors are mostly in the **liver**

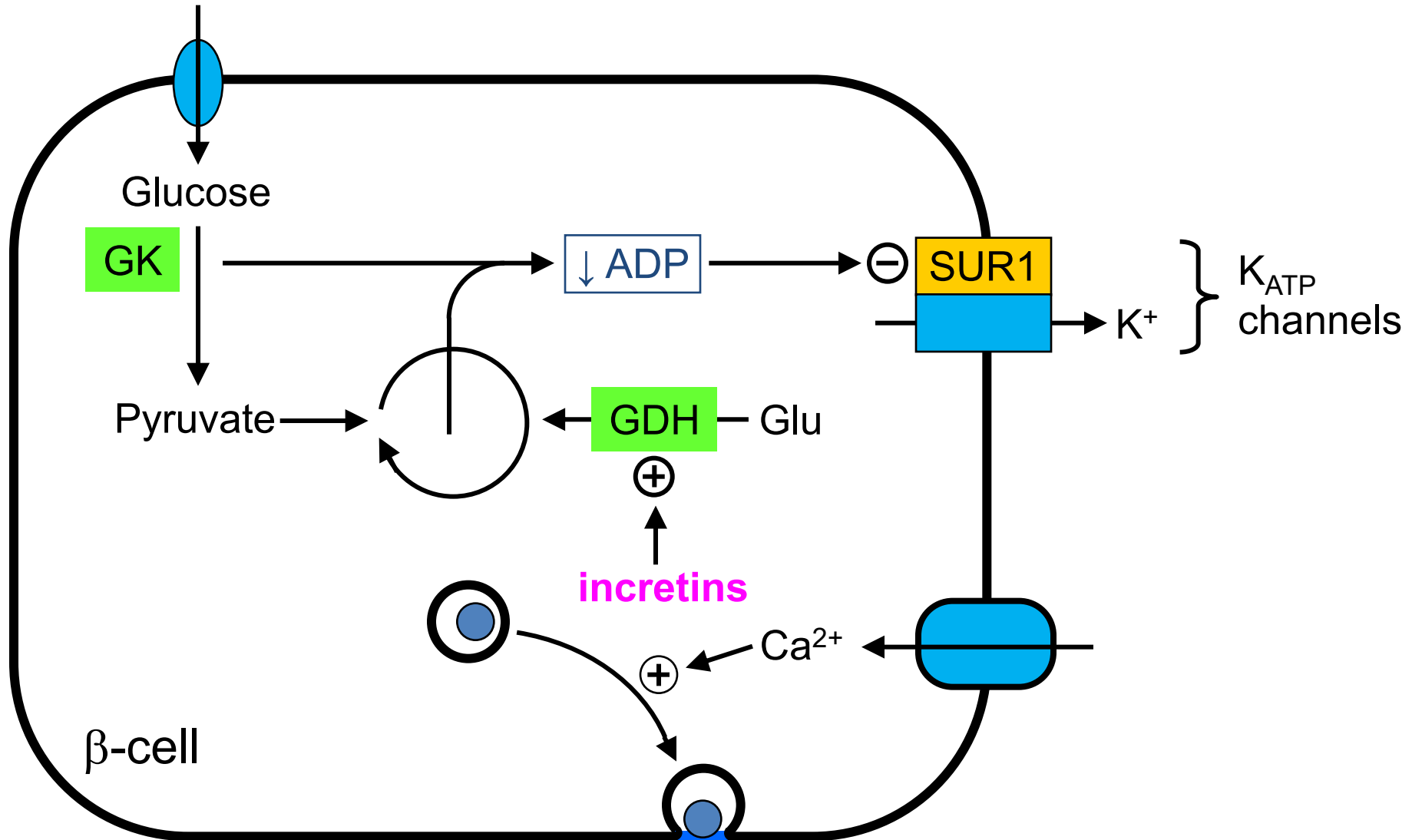
GIP receptors and GLP receptors are on **β-cells**, in **stomach**, in **brain**

Physiologically, glucagon receptors and GLP receptors show no cross-reactivity

A mixed meal stimulates GLP-1 secretion



How do incretins stimulate insulin secretion?



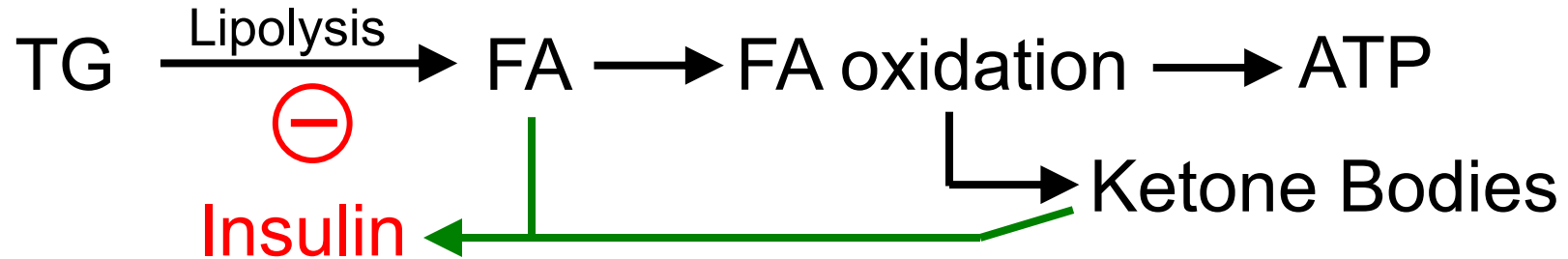
Incretins enhance glucose-induced insulin secretion

Incretins stimulate insulin secretion only at [glucose] greater than 100 mg/dl

Incretins delay gastric emptying and promote satiety

DPP-4 degrades incretins

Fatty acids and ketone bodies also enhance glucose-induced insulin secretion



**Insulin prevents the development of ketoacidosis
(A person who has a functioning endocrine
pancreas does not develop ketoacidosis.)**

ScholarRx 2800.18

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Summary

High serum glucose stimulates insulin production by pancreatic beta cells.

Insulin inhibits lipolysis and endogenous glucose production (glycogenolysis, gluconeogenesis) and stimulates glucose utilization (glycolysis, glycogen synthesis).

Insulin is produced from a precursor polypeptide. Endogenous insulin can be distinguished from synthetic (exogenous) insulin by the presence of the connecting (C) peptide that is processed out of mature insulin in secretory granules

Insulin action is opposed by glucagon, epinephrine and cortisol through stimulation of lipolysis, glycogenolysis and gluconeogenesis

Incretins (GIP, GLP) are released in response to mixed meals (especially dietary amino acids and fatty acids); they delay gastric emptying, promote satiety and increase glucose-dependent insulin release